INCH-POUND

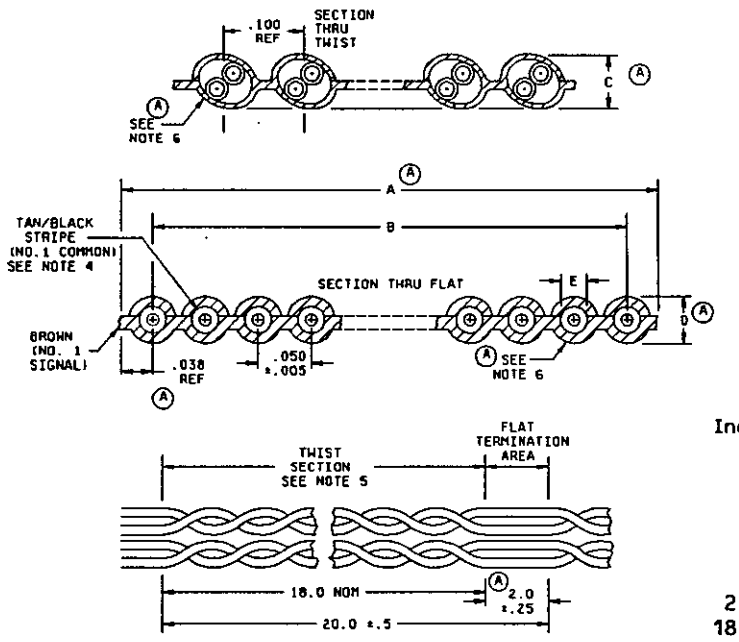
MIL-C-49055/15A
11 February 1993
SUPERSEDING
MIL-C-49055/15
16 August 1989

MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, FLEXIBLE, FLAT, UNSHIELDED, ROUND CONDUCTOR, COLOR CODED, STRANDED, 150°C, MULTIPLE TWISTED PAIRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49055.



Inches	mm
.005	0.13
.038	0.97
.050	1.27
.100	2.54
.25	6.4
.50	12.7
2.0	50.
18.0	457.
20.0	508.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All undimensioned pictorial representations are for references purpose only.
4. Red striping on tan wire (no.1 common) is required only on tan wire next to number 1 (brown) wire.
5. The transition area is included in twisted section to assure a full 2 inches (50.8 mm) of .050 inch (1.27 mm) conductor spacing in the termination area.
(A) 6. The laminate or substrate may be single or double sided.

FIGURE 1. Flat cable, twisted pairs with flat termination sections.

(A) denotes changes.

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(A) TABLE I. Dash numbers and description for figure 1. 1/

Dash number	AWG (conductor stranding)	Number of pairs	Dimensions					Weight (lbs/1000 ft.)	
			A max	B	C max	D ±0.003	E nom O.D. 2/	Nom	Max
01	28 (7x36)	5	0.541	0.450±0.015	0.090	0.041	0.035	18.10	19.01
02	28	7	0.741	0.650±0.015	0.090	0.041	0.035	25.34	26.61
03	28	8	0.846	0.750±0.015	0.090	0.041	0.035	28.96	30.41
04	28	10	1.046	0.950±0.015	0.090	0.041	0.035	36.20	38.01
05	28	13	1.346	1.250±0.015	0.090	0.041	0.035	47.06	49.41
06	28	17	1.751	1.650±0.015	0.090	0.041	0.035	61.54	64.62
07	28	18	1.851	1.750±0.015	0.090	0.041	0.035	65.16	68.42
08	28	20	2.051	1.950±0.020	0.090	0.041	0.035	72.40	76.02
09	28	25	2.551	2.450±0.020	0.090	0.041	0.035	90.50	95.03
10	28	30	3.051	2.950±0.020	0.090	0.041	0.035	108.60	114.03
11	28	32	3.251	3.150±0.020	0.090	0.041	0.035	115.84	121.63
12	26 (7x34)	5	0.541	0.450±0.015	0.095	0.045	0.039	23.84	25.03
13	26	7	0.741	0.650±0.015	0.095	0.045	0.039	31.78	33.29
14	26	8	0.846	0.750±0.015	0.095	0.045	0.039	36.78	38.14
15	26	10	1.046	0.950±0.015	0.095	0.045	0.039	45.40	47.67
16	26	13	1.346	1.250±0.015	0.095	0.045	0.039	59.02	61.97
17	26	17	1.751	1.650±0.015	0.095	0.045	0.039	77.18	81.04
18	26	18	1.851	1.750±0.015	0.095	0.045	0.039	81.72	85.81
19	26	20	2.051	1.950±0.020	0.095	0.045	0.039	90.80	95.34
20	26	25	2.551	2.450±0.020	0.095	0.045	0.039	113.50	119.18
21	26	30	3.051	2.950±0.020	0.095	0.045	0.039	136.20	143.01
22	26	32	3.251	3.150±0.020	0.095	0.045	0.039	145.28	152.54

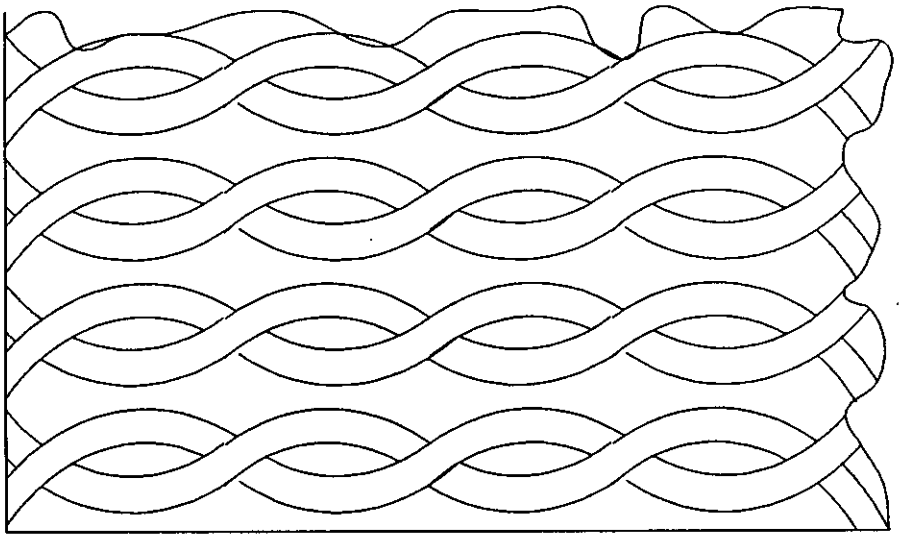
1/ Nominal values are for information only, not requirements.
2/ The minimum conductor insulation thickness shall be .007 inch.

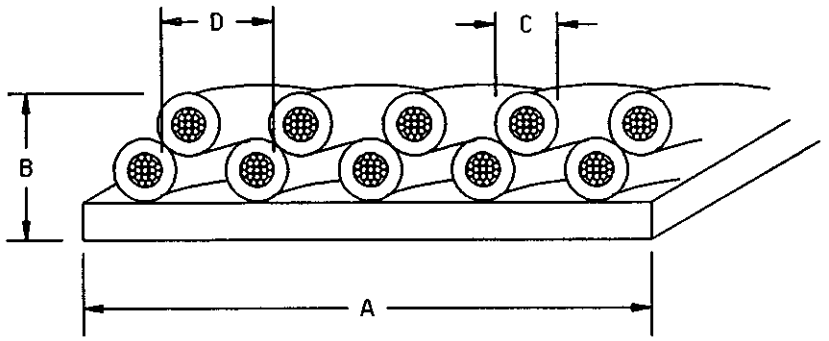
Inches	mm
.015	0.38
.020	0.51
.035	0.89
.039	0.99
.041	1.04
.045	1.14
.090	2.29
.095	2.41
.450	11.43

Inches	mm
.541	13.74
.650	16.51
.741	18.82
.750	19.05
.846	21.49
.950	24.13
1.046	26.57
1.250	31.75
1.346	34.19

Inches	mm
1.650	41.91
1.750	44.45
1.751	44.48
1.851	47.02
1.950	49.53
2.051	52.10
2.551	64.80
3.051	77.50
3.150	80.01
3.251	82.58

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NOTES:
1. Dimensions are in inches.
2. All undimensional pictorial representations are for reference purposes only.
3. Wires shall be removable from laminate without causing any degradation to the insulation of the wire.
4. This cable is recommended for discrete wiring but may be used with insulation displacement connectors.
5. Twisted pairs are shown offset to represent possible position of twist lay.

(A) FIGURE 2. Flat cable, twisted pairs with removable laminate.

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(A) TABLE II. Dash numbers and description for figure 2. 1/

Dash number	AWG (conductor stranding)	Number of pairs	Dimensions 2/				Weight (lbs./1000 ft.)	
			A (max)	B (max)	C (nom) 3/	D (ref)	Nominal	Maximum
23	28(7x36)	5	.330	.075	.032	.062	13.6	15.1
24	28	7	.448	.075	.032	.062	19.0	21.1
25	28	8	.512	.075	.032	.062	21.8	24.1
26	28	10	.640	.075	.032	.062	27.2	30.2
27	28	13	.832	.075	.032	.062	35.3	39.2
28	28	17	1.080	.075	.032	.062	46.2	51.3
29	28	18	1.160	.075	.032	.062	48.9	54.3
30	28	20	1.280	.075	.032	.062	54.4	60.3
31	28	25	1.600	.075	.032	.062	68.0	75.4
32	28	30	1.920	.075	.032	.062	81.6	90.5
33	28	32	2.050	.075	.032	.062	87.0	96.5
34	26(7X34)	5	.370	.080	.035	.070	17.8	19.3
35	26	7	.504	.080	.035	.070	24.9	27.0
36	26	8	.576	.080	.035	.070	28.4	30.8
37	26	10	.720	.080	.035	.070	35.6	38.6
38	26	13	.936	.080	.035	.070	46.2	50.1
39	26	17	1.224	.080	.035	.070	60.5	65.5
40	26	18	1.296	.080	.035	.070	64.0	69.4
41	26	20	1.440	.080	.035	.070	71.1	77.1
42	26	25	1.800	.080	.035	.070	88.9	96.4
43	26	30	2.160	.080	.035	.070	106.7	115.7
44	26	32	2.304	.080	.035	.070	113.8	123.4

1/ Nominal values are for information only, not requirements.
2/ Dimensions are in inches.
3/ The minimum conductor insulation thickness shall be .007 inch.

Inches	mm	Inches	mm	Inches	mm
.032	0.81	.504	12.80	1.280	32.51
.035	0.89	.512	13.01	1.296	32.92
.062	1.58	.576	14.63	1.440	36.57
.070	1.78	.640	16.26	1.600	40.64
.072	1.83	.720	18.289	1.800	45.72
.075	1.91	.832	21.13	1.920	48.77
.080	2.03	.936	23.77	2.050	52.07
.330	8.38	1.080	27.43	2.160	54.86
.370	9.40	1.160	29.46	2.304	58.52
.448	11.38	1.224	31.09		

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(A) REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2 and tables I and II.

Conductor insulation material: Cross-linked polyalkene/cross-linked polyvinyldene fluoride; ethlene tetrafluoroethylene (ETFE); ethylene tetrafluoroethylene modified cross-linked; ethylene-chloritrifluoroethylene (ECTFE); fluorinated ethylene propylene (FEP); polytetrafluoroethylene (PTFE); perfluoral koxy (PFA). NOTE: Laminate material may be the same as conductor insulation material, or may be any other compatible material listed herein.

Voltage rating: 300 volts rms at sea level.

Conductor coating: Bare or tin.

Color: Color coding for twist and flat configuration (see figure 1): Ten color repeat (brown, red, orange, yellow, green, blue, violet, gray, white, and black) with common ground as tan with black stripe (see note 4 of figure 1).

Color coding for twisted configuration only (see figure 2): Nine color repeat (black, brown, red, orange, yellow, green, blue, violet, and gray) with common ground as white.

Weight: See tables I and II.

Insulation flaws: 2,500 volts rms.

Dielectric withstanding voltage: 2,000 volts rms.

Insulation resistance: 500 megohms - 1,000 feet (minimum).

Conductor resistance at 23°C:

28 AWG: 69 ohms/1,000 ft. max.
26 AWG: 41 ohms/1,000 ft. max.

High temperature aging: Flat section (for figure 1 configuration only): Aging temperatures to be in accordance with table Ia of MIL-C-49055 using a .440 (11.18 mm) inch mandrel. Twisted section (for figures 1 and 2): Aging temperatures to be in accordance with table Ia of MIL-C-49055 using .880 (20.57 mm) inch mandrel.

Temperature rating: -55°C to 150°C.

Thermal shock: -55°C -5°C to 150°C +5°C, shrinkage .0625 inch maximum either end.

Flexing endurance: -55°C -5°C; 150°C +5°C, 300 cycles for 26 AWG, 500 cycles for 28 AWG; to be in accordance with MIL-C-49055 with the following exceptions:

a. The length between clamping devices shall be 2.00 (50.8 mm) to 2.05 inches (52.1 mm).

b. The cable length shall be 4.50 (114.3 mm) to 4.55 inches (115.57 mm).

c. The clamping device radius shall be .440 inch (11.18 mm) typical.

Moisture resistance:

Insulation resistance: 50 megohms/1,000 ft. minimum.

Low pressure (elevated temperature): 150°C +5°C maximum conductor temperature.

Tear groove propagation: Not applicable.

Fungus resistance: To be in accordance with MIL-C-49055. As an option, the insulation and laminate material shall be certified as being "non nutrient" to fungus growth.

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Pair lay length: .5 inch nominal lay.

Number of pairs: See tables I and II.

Laminate or substrate: .005 inch minimum thickness.

Part or Identifying Number (PIN): M49055/15- (dash number from table I or II) ("T" for tin plating and blank for bare copper) Example: M49055/15-11T and M49055/15-11.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC, OS, SH
Air Force - 17, 99
DLA - IS, ES
NSA
NAS
NASA - NA

User activities:
Army - AT, ME
Navy - MC

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-2003-2)

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